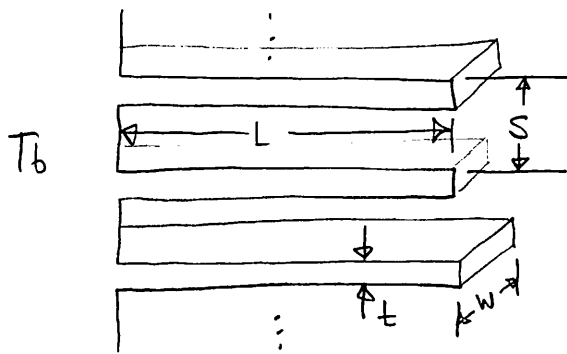


Overall Surface Efficiency (η_o)

The overall surface efficiency characterizes the performance of an array of fins.

$$\eta_o = \frac{q_t}{q_{\max}} = \frac{q_t}{h A_t \theta_b}$$

$q_t \equiv$ total heat transfer of fins & base of area (A_t)



Assuming:

- 1) $N \equiv$ number of fins in array
- 2) $A_f \equiv$ exposed area per fin
- 3) $A_b \equiv$ base area between fins
- 4) $\eta_f \equiv$ individual fin efficiency

$$A_t = N A_f + A_b$$

$$q_t = N \eta_f h A_f \theta_b + h A_b \theta_b$$

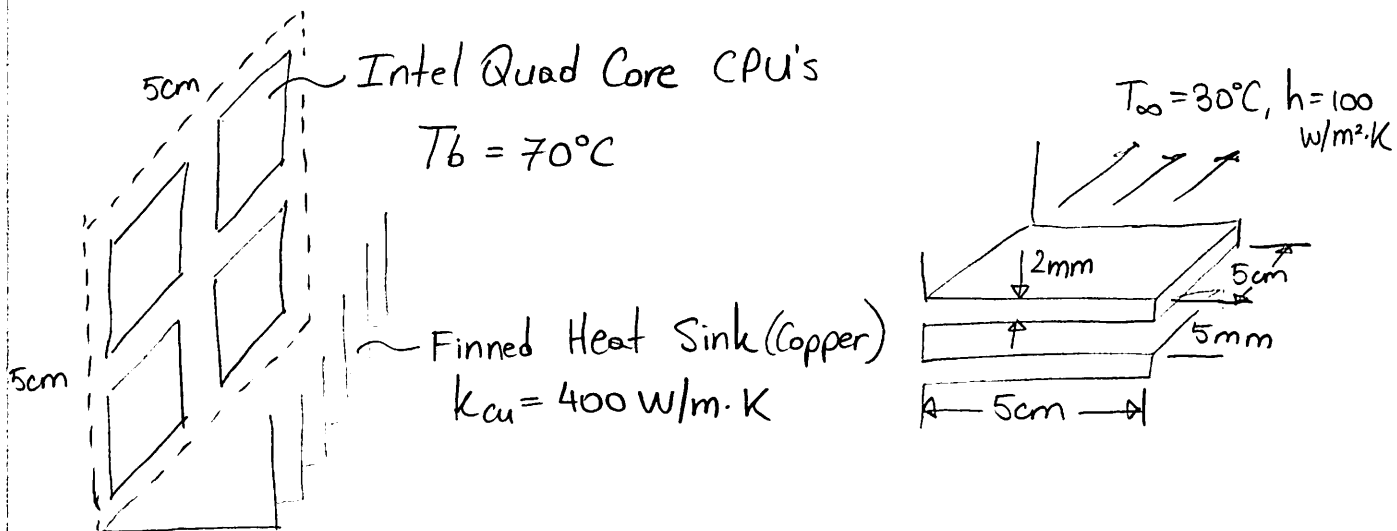
$$q_t = h [N \eta_f A_f + (A_t - N A_f)] \theta_b = h A_t \left[1 - \frac{N A_f}{A_t} (1 - \eta_f) \right] \theta_b$$

$$\eta_o = 1 - \frac{N A_f}{A_t} (1 - \eta_f) \Rightarrow \text{Overall surface efficiency}$$

We can now also calculate the thermal resistance of a fin array.

$$R_o = \frac{\theta_b}{q_t} = \frac{1}{h A_t \eta_o} \Rightarrow \text{Thermal resistance of a fin array.}$$

Example / CPU cooling



What is the increase in heat transfer due to the use of fins?

Due to the small fin thickness, we can ignore the fin edges, and use 1D fin analysis.

$$A_f = 2w \left(L + \frac{t}{2} \right) = 2(0.05\text{m}) \left(0.05\text{m} + \frac{0.002\text{m}}{2} \right) = 0.0051\text{m}^2$$

$$A_t = \underbrace{N(A_f)}_{\text{Total fin Area}} + \underbrace{A_b}_{\text{Total Base Area}} = \left[0.0051\text{m}^2 + (0.003\text{m})(0.05\text{m}) \right] \underbrace{\left[\frac{0.05\text{m}}{0.005\text{m}} \right]}_N$$

$$A_t = 0.0525\text{m}^2$$

But wait, we didn't check the Biot #! $Bi_t = \frac{h t}{k} = \frac{(100)(0.002)}{400}$

We know for our individual fin:

$$Bi_t = 5 \times 10^{-4} \ll 0.1$$

$$\eta_{fin} = \frac{\tanh(mL)}{mL} = \frac{\tanh\left(\sqrt{\frac{hP}{kA}} L\right)}{\sqrt{\frac{hP}{kA}} \cdot L} = \tanh\left(\frac{\sqrt{\frac{2(0.05)(100)}{(400)(0.05)(0.002)}} (0.05)}{\dots}\right) \quad \text{OK!}$$

$$\eta_o = 1 - \frac{N A_f}{A_t} (1 - \eta_f) = 1 - \frac{10(0.0051)}{0.0525} (1 - 0.833) = 0.84$$

$$q_o = h A_o \theta_b = (100)(0.0525)(70 - 30)(0.84) = 176.4\text{W}$$

$$q_o = h A_o \theta_b = (100)(0.05)(0.002)(40) = 10\text{W}$$

Steady Multi-Dimensional Heat Transfer (Shape Factor)

If a problem is steady (ss) and has isothermal surfaces, we can define what's called a "shape factor"

$$Q = Sk\Delta T$$

↳ Obtained analytically or numerically

$$R_{th} = \frac{1}{kS}$$

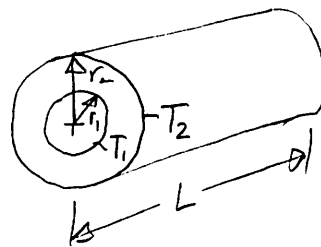
For example, let's see how it works out for a plane wall:

$$R_{wall} = \frac{L}{kA} = \frac{1}{kS} \Rightarrow \boxed{S_{wall} = \frac{A}{L}}$$

$$\boxed{Q = kA \frac{\Delta T}{L} = kA \frac{\partial T}{\partial x}} \Rightarrow \text{Makes sense}$$

How about a cylinder:

$$R_{cyl} = \frac{\ln(r_2/r_1)}{2\pi kL}$$



$$\boxed{S_{cyl} = \frac{1}{kR_{cyl}} = \frac{2\pi L}{\ln(r_2/r_1)}}$$

⇒ Very simple and in general not needed for simple geometries as R_{th} is available & our previous analysis holds.

However for more complex shapes, it is a very useful concept.

Table 5.4 Conduction shape factors: $Q = S k \Delta T$, $R_t = 1/(kS)$.

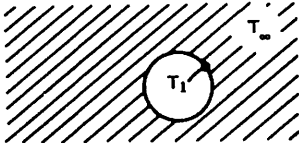
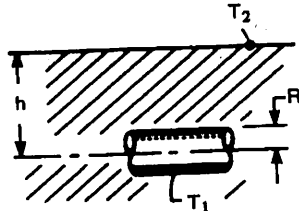
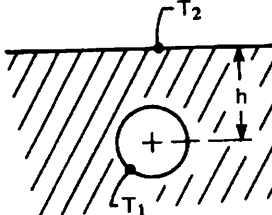
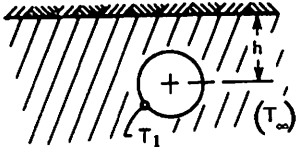
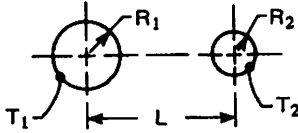
Situation	Shape factor, S	Dimensions	Source
1. Conduction through a slab	A/L	meter	Example 2.2
2. Conduction through wall of a long thick cylinder	$\frac{2\pi}{\ln(r_o/r_i)}$	none	Example 5.9
3. Conduction through a thick-walled hollow sphere	$\frac{4\pi(r_o r_i)}{r_o - r_i}$	meter	Example 5.10
4. The boundary of a spherical hole of radius R conducting into an infinite medium 	$4\pi R$	meter	Problems 5.19 and 2.15
5. Cylinder of radius R and length L , transferring heat to a parallel isothermal plane; $h \ll L$ 	$\frac{2\pi L}{\cosh^{-1}(h/R)}$	meter	[5.16]
6. Same as item 5, but with $L \rightarrow \infty$ (two-dimensional conduction)	$\frac{2\pi}{\cosh^{-1}(h/R)}$	none	[5.16]
7. An isothermal sphere of radius R transfers heat to an isothermal plane; $R/h < 0.8$ (see item 4) 	$\frac{4\pi R}{1 - R/2h}$	meter	[5.16, 5.17]

Table 5.4 Conduction shape factors: $Q = S k \Delta T$, $R_t = 1/(kS)$ (con't).

Situation	Shape factor, S	Dimensions	Source
8. An isothermal sphere of radius R, near an insulated plane, transfers heat to a semi-infinite medium at T_∞ (see items 4 and 7) 	$\frac{4\pi R}{1 + R/2h}$	meter	[5.18]
9. Parallel cylinders exchange heat in an infinite conducting medium 	$\frac{2\pi}{\cosh^{-1} \left(\frac{L^2 - R_1^2 - R_2^2}{2R_1 R_2} \right)}$	none	[5.6]
10. Same as 9, but with cylinders widely spaced; $L \gg R_1$ and R_2	$\frac{2\pi}{\cosh^{-1} \left(\frac{L}{2R_1} \right) + \cosh^{-1} \left(\frac{L}{2R_2} \right)}$	none	[5.16]
11. Cylinder of radius R_i surrounded by eccentric cylinder of radius $R_o > R_i$; centerlines a distance L apart (see item 2)	$\frac{2\pi}{\cosh^{-1} \left(\frac{R_o^2 + R_i^2 - L^2}{2R_o R_i} \right)}$	none	[5.6]
12. Isothermal disc of radius R on an otherwise insulated plane conducts heat into a semi-infinite medium at T_∞ below it	$4R$	meter	[5.6]
13. Isothermal ellipsoid of semimajor axis b and semiminor axes a conducts heat into an infinite medium at T_∞; $b > a$ (see 4)	$\frac{4\pi b \sqrt{1 - a^2/b^2}}{\tanh^{-1} \left(\sqrt{1 - a^2/b^2} \right)}$	meter	[5.16]